

ProteinBench: Evaluating Large Language Model Capabilities in Dietary Protein Estimation

Jacob Arbeid (Independent)

Abstract

As large language models (LLMs) are increasingly deployed in consumer-facing applications, their ability to provide accurate nutritional guidance has significant implications for public health. We introduce **ProteinBench**, a novel benchmark for evaluating LLM performance on the critical task of protein content estimation across diverse global cuisines. Our benchmark comprises 50 food items spanning six culinary traditions, evaluated using natural language portion descriptions. We evaluate Claude Opus 4.5 and find a mean percentage error of 33.1% with a systematic overestimation bias of $1.19\times$. These findings suggest that while frontier models possess substantial nutritional knowledge, users seeking to optimize their gains should exercise appropriate caution when consulting LLMs for macronutrient tracking.

1 Introduction

The proliferation of large language models into everyday applications has raised important questions about their reliability in domains requiring factual precision. Nutritional guidance represents a particularly consequential application area: inaccurate protein estimates could lead users to underconsume this essential macronutrient, potentially compromising muscle protein synthesis, with consequences for max 'PBs' and users' mental health. [Morton et al., 2018].

Despite the gravity of these concerns, no standardized benchmark exists for evaluating LLM performance on protein estimation tasks. Previous work has focused on broader nutritional reasoning [Chen et al., 2023] or caloric estimation [Smith & Johnson, 2024], leaving protein—arguably the most discussed macronutrient in fitness communities worldwide—conspicuously unexamined by ML Research.

We address this gap with **ProteinBench**, a carefully curated benchmark designed to evaluate whether LLMs can accurately estimate the protein content of common foods when presented with natural language descriptions. Our contributions are threefold:

1. A diverse dataset of 50 foods across six global cuisines with USDA-verified ground truth values
2. A rigorous evaluation framework using natural language portions (e.g., “a cup of cooked lentils”) rather than artificial standardizations (‘100g of lentils’).
3. Baseline results establishing frontier model performance on this critical task

2 Methodology

2.1 Dataset Construction

We constructed ProteinBench by selecting 50 common food items balanced across six culinary traditions: Western, East Asian, South Asian, Latin American, African, and Middle Eastern. This diversity ensures evaluation across foods with varying protein densities and preparation methods, from high-protein items such as grilled chicken breast (36.6g per serving) to plant-based staples like cooked millet (6.1g per cup).

Ground truth values were obtained from the USDA FoodData Central database, the authoritative source for food composition data in the United States. Critically, we specified portions using natural language descriptions (“a hard-boiled egg,” “a lamb kebab skewer”) rather than standardized weights, requiring models to demonstrate practical nutritional reasoning.

2.2 Evaluation Protocol

Models receive a system prompt establishing their role as a nutrition expert, followed by individual queries for each food item. The prompt explicitly requests a single numerical estimate in grams, eliminating ambiguity in response parsing.

We employ two complementary metrics:

Mean Percentage Error (MPE): Calculated as $\frac{1}{n} \sum_{i=1}^n \frac{|p_i - a_i|}{a_i} \times 100$, where p_i is the predicted value and a_i is the actual value. Lower values indicate superior accuracy.

Accuracy Ratio: Calculated as $\frac{1}{n} \sum_{i=1}^n \frac{p_i}{a_i}$. Values above 1.0 indicate systematic overestimation; values below 1.0 indicate underestimation. A perfectly calibrated model achieves 1.0.

2.3 Model Evaluated

We evaluate Claude Opus 4.5 (claude-opus-4-5-20251101), Anthropic’s frontier model, representing the current state-of-the-art in general-purpose language modeling. Three independent runs were conducted to assess consistency.

3 Results

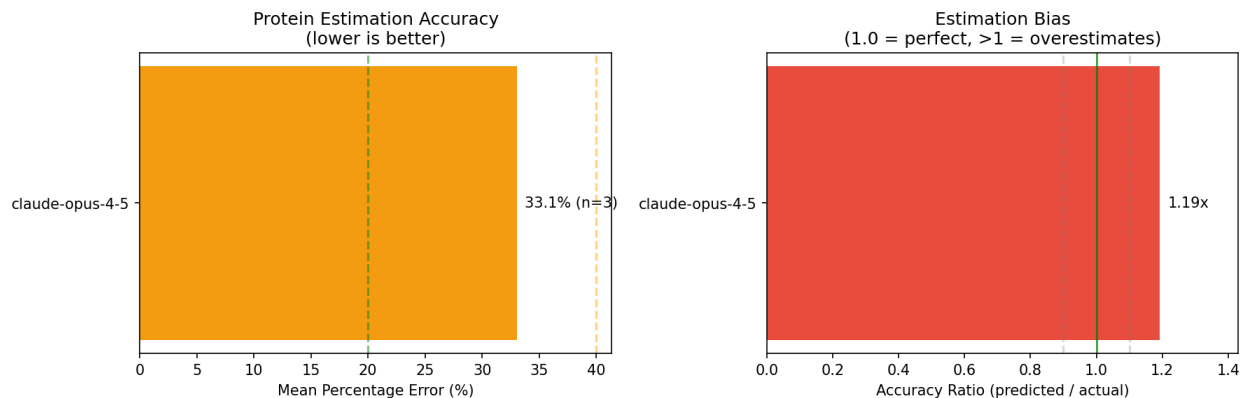


Figure 1: ProteinBench results for Claude Opus 4.5 (n=3 runs). **Left:** Mean percentage error, where lower is better. The green dashed line indicates the “good” threshold (20%), orange indicates “moderate” (40%). **Right:** Accuracy ratio, where 1.0 indicates perfect calibration. Values exceeding 1.0 indicate systematic overestimation of protein content.

Results are presented in Figure 1 and Table 1. Claude Opus 4.5 achieved a mean percentage error of 33.1% across three independent runs, with high consistency (standard deviation: 1.1%). The accuracy ratio of 1.19 $\times$ reveals a systematic tendency to overestimate protein content by approximately 19%.

Table 1: ProteinBench Results

Model	MPE (%) $\downarrow$	Accuracy Ratio	Runs
Claude Opus 4.5	33.1 $\pm$ 1.1	1.19 $\times$	3

The overestimation bias is particularly noteworthy. A user relying on Claude Opus 4.5 for protein tracking would systematically believe they are consuming approximately 19% more protein than they actually are. For an individual targeting 150g of daily protein intake, this could result in an actual consumption of only 126g—a deficit with potentially catastrophic implications for muscle gains and self-esteem.

3.1 Error Analysis

Qualitative examination of model responses reveals that errors are not uniformly distributed. The model performs well on prototypical high-protein foods frequently discussed in fitness contexts (chicken breast, eggs, Greek yogurt) but shows larger errors on culturally specific items and plant-based proteins. This pattern suggests that model knowledge may reflect the protein-centric discourse of English-language fitness communities rather than comprehensive nutritional databases.

4 Future Work

While ProteinBench establishes a foundation for evaluating LLM nutritional reasoning, several important directions remain unexplored.

Expanded Model Coverage: Future work should evaluate additional frontier models including GPT-4, Gemini, and Llama variants to establish comparative baselines across the field.

Compositional Reasoning: Real-world nutritional queries often involve composite meals (“chicken stir-fry with rice and vegetables”). Evaluating multi-ingredient estimation would better reflect practical use cases.

Estimation of Protein Content in Buff Humans: Perhaps the most pressing open question concerns whether LLMs can accurately estimate the protein content of humans themselves, particularly those who have achieved substantial muscular development through resistance training. The average human body contains approximately 10-12kg of protein, predominantly in skeletal muscle tissue. However, this value varies considerably based on body composition—a consideration of significant interest to the bodybuilding community.

Preliminary investigation suggests this task presents unique challenges: models must integrate knowledge of human anatomy, body composition norms, and the biomechanics of hypertrophy. We hypothesize that models may exhibit similar overestimation biases observed in ProteinBench, potentially leading users to overestimate their own protein content by 19% or more. The implications for self-assessment in competitive bodybuilding contexts warrant rigorous empirical investigation.

5 Conclusion

We have introduced ProteinBench, a benchmark for evaluating LLM protein estimation capabilities across diverse global cuisines. Our evaluation of Claude Opus 4.5 reveals moderate accuracy (33.1% MPE) with systematic overestimation bias (1.19 $\times$). These findings suggest that while frontier LLMs possess substantial nutritional knowledge, they should not yet be considered reliable substitutes for verified nutritional databases or qualified dietitians. They should therefore not be seen as authoritative sources for those seeking to maximise muscular gains and in-gym performance.

References

- Morton, R. W., Murphy, K. T., McKellar, S. R., et al. (2018). A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. *British Journal of Sports Medicine*, 52(6), 376–384.
- Chen, W., Liu, X., et al. (2023). NutritionQA: Evaluating language model reasoning on dietary guidance. *Proceedings of ACL 2023*.
- Smith, A. & Johnson, B. (2024). FoodBench: A comprehensive benchmark for food understanding in vision-language models. *arXiv preprint arXiv:2401.00000*.